

Summary. AxonOS is a Rust microkernel for closed-loop brain–computer interfaces with analytical schedulability proofs, an open application–binary interface, and a pre-registered Phase-1 measurement protocol scheduled for Q2 2026. It addresses a structural gap: closed-loop BCI software currently lacks shared infrastructure for validated real-time safety claims, and no general-purpose RTOS provides analytically schedulable, **unsafe-free**, capability-isolated foundations for the domain. The work is open source under Apache-2.0 / MIT (code) and CC-BY-SA-4.0 (specifications), preprint v19 on arXiv–route to submission, and is in Phase 0 of a defined 24-month trajectory.

Problem

Closed-loop neurostimulation operates on a 4-millisecond budget. Every acquisition epoch must be filtered, classified, and converted to a stimulation command before the next epoch arrives. Missing one epoch is not a performance regression. Under IEC 60601-1, which governs medical electrical equipment, a missed deadline in a closed control loop is a violation of *essential performance*—the same regulatory category that covers pacemakers and infusion pumps.

The software substrate in those adjacent fields meets that category through a specific combination of disciplines: small kernels with analytically provable scheduling guarantees, written in languages or verified by methods that match the consequence of failure, on hardware whose timing properties are documented to the cycle. Pacemakers run on certified RTOSes with formal verification artifacts. Avionics runs on DO-178B and ARINC-653 partitioned systems.

Brain-computer interfaces in 2026, almost without exception, do not. The most ambitious BCI research in the world—academic, clinical, and commercial—runs on Linux, Windows, hobbyist real-time operating systems, or one-off proprietary firmware. None of these provide an analytical schedulability proof in a language that eliminates memory safety defects at compile time. None enforces capability constraints on neural-data egress at the kernel boundary. None publishes its application–binary interface as a versioned specification under an open licence.

This is a structural gap, not a product gap. It is the gap between an industry that builds demonstrations and an industry that ships regulated medical devices. The same gap existed in avionics in the 1970s and was closed by shared standards over the subsequent two decades. The work proposed here is the first step of an analogous closure for BCI.

Approach

AxonOS is a `#![no_std]#![forbid(unsafe_code)]` Rust microkernel for Cortex-M class microcontrollers. Five elements distinguish its construction:

Analytical schedulability. The pipeline task set is admitted under a Liu–Layland EDF test with a conservative ceiling $U_{\max} = 0.25$. Worst-case execution times are derived from the ARM Cortex-M4 Technical Reference Manual r0p1 and the STM32F407 datasheet, not from runtime profiling. Six identified sources of timing pessimism (S1–S6) are quantified, and the gap between cold-path WCET (663 μ s) and the engineering budget WCET (818 μ s) is disclosed as a 23 % engineering allowance, not asserted as a sound bound. The path to closing it (aiT static analysis under the AHB bus-matrix timing model) is specified in the Phase-1 protocol.

Structural data minimisation. Application code receives only typed, rate-limited, attested intent observations. Raw EEG, continuous emotion inference, and cognitive profile reads are absent from the kernel’s type catalogue, not denied by runtime check. An information-theoretic mutual-information bound on application-observable neural data is proved against an active attacker model. This converts what is conventionally a policy claim into a structural one.

Memory-ordering correctness. The kernel’s single producer/single consumer ring buffer—the only **unsafe** surface in the kernel—is verified by the Kani bounded model checker for Release/Acquire correctness on the Cortex-M4 memory model. Three harnesses are delivered and pass within stated loop bounds.

Open application-binary interface. The intent wire format and capability bitfield are specified in RFC-0006 under CC-BY-SA-4.0, with conformance test vectors. Any implementation that emits compliant bytes is AxonOS-compatible. This eliminates vendor lock-in at the most important interface in the system and makes “AxonOS-compatible” a verifiable property of the wire, not a marketing claim.

Pre-registered falsification. Five numbered predictions (P1–P5) state in advance what a Phase-1 measurement study must find for this analysis to be confirmed or refuted. Falsification thresholds are fixed before measurement. The replication artefacts are pinned at the same repository commit as the preprint.

Status and near-term milestones

Now (Phase 0). Preprint v19 on arXiv-route to submission (analytical specification, no measurement claims from our own hardware). Six engineering RFCs published under CC-BY-SA-4.0. SDK v0.4.0 security-audited and released (14 audit items resolved). Memorandum of understanding with a US Northeast ALS rehabilitation centre covering 200+ patients, signed Q1 2026.

Q2 2026 (Phase 1). GPIO-instrumented WCRT measurement on STM32H573 reference fixture with Saleae Logic Pro 16 at 10 ns resolution. Cross-board validation on $n \geq 3$ STM32F407 units. Direct power measurement on the reference board. Promotion of RFC-0006 ABI from draft to active status based on validated wire format. The result will be published regardless of outcome, including if P1–P5 are refuted.

Q3–Q4 2026 (Phase 2). First 8-channel clinical kit deployment with the ALS partner. Online closed-loop classifier performance reported alongside offline benchmark. Second clinical site scoping.

2027 (Phase 3). FDA Pre-Submission (Q-Sub) filing. Ferrocene-qualified toolchain integration. ISO 14971 risk management file complete. Second arXiv preprint covering Phase-1 measurement results.

Risks

Phase-1 refutes the analytical predictions. This is the principal scientific risk. P1–P5 are written so that a single number above a stated threshold refutes the prediction. The analytical pessimism budget of 23 % is engineering judgment, not a sound bound; if the unmodelled effects exceed this allowance, the schedulability margin against $U_{\max}$ disappears. Mitigation: falsification thresholds are public and the result will be published either way, preserving credibility regardless of outcome. A refuted prediction redirects engineering, it does not end the project.

Solo-founder execution risk. The technical depth is high (real-time scheduling, embedded Rust, formal methods, information theory, BCI domain) and currently rests on one engineer. Mitigation: scientific advisory expansion in Q3 2026; explicit identification of a co-founder or VP Engineering search as a pre-Series-A condition.

Regulatory timeline slip. FDA Q-Sub filing in 2027 depends on Phase 2 clinical pilot data quality and on notified-body engagement that has not yet started. Mitigation: IEC 62304 preliminary mapping is in preprint v19 as a self-assessment, not a notified-body declaration; the gap to formal compliance is enumerated.

Adjacent-platform competition. A general-purpose RTOS vendor (Wind River, SYSGO, Green Hills) could in principle ship a BCI-specific safety case on top of an existing certified kernel. This is the most plausible substitutive threat. Mitigation: AxonOS is published as open-source; substitutive

competition that closes the source surrenders the reproducibility advantage that pre-clinical validation requires. The competitive position is not technical exclusivity, it is open-substrate positioning.

Adoption by larger labs. Academic groups with established Linux-based pipelines may resist re-platforming. Mitigation: the SDK is designed to be a bolt-on for existing pipelines (capability-typed event stream, not a full replacement of signal processing), and the ABI specification allows adoption to start at the interface boundary, not at full kernel deployment.

Engineering principles

These five rules govern technical decisions and are visible in every artifact produced by the project:

- 1. No claim above its evidence level. Measurements on one board for 12 hours are reported as runtime-measured (Level 2 in the taxonomy of RFC-0003), not as validated.
- 2. No **unsafe** in reviewable modules. Hardware register access is confined to audited Peripheral Access Crates; the rest is `#![forbid(unsafe_code)]`.
- 3. No heap allocation on the hot path. Static buffers, sized at compile time, sized to fit the WCET budget.
- 4. No silent recovery from inconsistent state. Poisoned mutexes, clock violations, and protocol mismatches surface as errors, not as defaults.
- 5. No proprietary lock-in via the kernel. The application-binary interface is published as an RFC under CC-BY-SA-4.0. Third-party implementations are welcomed.

These are not aspirational. They are how decisions are made, and they are visible in the source code, the RFCs, and the preprint.

AxonOS is not a complete answer. It is a foundation on which the answer can be built, in the open, by parties who agree on a minimum standard of analytical rigor. This document describes that foundation and the work remaining. All quantitative claims are verifiable against preprint v19 (May 2026) and the linked specifications. Independent replication is encouraged and will be incorporated with attribution.

Verifiable artifacts

Preprint v19	Analytical specification, 14 pages, 53 references (arXiv pending)
Engineering RFCs	github.com/AxonOS-org/axonos-rfcs (RFC-0001 through RFC-0006, CC-BY-SA-4.0)
SDK	github.com/AxonOS-org/axonos-sdk (v0.4.0, Apache-2.0 OR MIT, security audit complete)
Consent protocol	github.com/AxonOS-org/axonos-consent (v0.4.0, reference implementation)
Site	axonos.org
Writing	medium.com/@AxonOS (38+ engineering essays since 2025)
Contact	connect@axonos.org (response within 5 business days to scoped enquiries)

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